

Counting Oriented Cycles Up to Dihedral Symmetry

March 10, 2026

Abstract

I count directed cycles on n edges under dihedral symmetry using a small transfer matrix coupled with Burnside's lemma. The adjacency rule forbids like-signed nonzero vertex signatures from being adjacent. A 6-state transfer matrix produces closed-walk counts; Burnside aggregates rotation and reflection fixed points to yield exact orbit counts. A full example for $n = 8$ reproduces the program values 288, 64, and 22. Optional primitivity and forbidden-subsequence filters follow immediately. Small n results match brute force and OEIS A053656.

1 Model

A cycle is encoded by a binary edge sequence $(e_0, \dots, e_{n-1}) \in \{0, 1\}^n$. Edge i connects vertices i and $i + 1 \pmod{n}$. Bit 0 orients clockwise, bit 1 counterclockwise. At vertex v I define the signature

$$\sigma(v) = 2 \cdot (\text{incoming edges at } v) - 2 \in \{-2, 0, +2\}.$$

Adjacency rule. Nonzero signatures of the same sign may not be adjacent around the cycle.

2 Symmetry

The dihedral group D_n (n rotations, n reflections; size $2n$) acts on edge sequences. A reflection reverses order and flips bits $0 \leftrightarrow 1$ to preserve geometric direction. Two sequences are equivalent if related by an element of D_n . Counting distinct cycles means counting orbits of this action.

3 Transfer matrix

Local state $s = (v, e)$ records previous vertex signature $v \in \{-2, 0, +2\}$ and current edge bit $e \in \{0, 1\}$; six states total. Given current edge e and next edge e' , the next signature is

$$\sigma' = 2(\mathbf{1}_{e=0} + \mathbf{1}_{e'=1}) - 2.$$

I admit a transition $(v, e) \rightarrow (\sigma', e')$ only if the adjacency rule holds for v, σ' . The resulting 6×6 matrix M has entries 1 for allowed transitions, 0 otherwise. The number of valid length- m closed walks is

$$C(m) = \text{tr}(M^m).$$

4 Burnside aggregation

Rotations. A rotation by k fixes sequences with period $d = \gcd(n, k)$. Grouping rotations by d yields

$$\text{RotSum}(n) = \sum_{d|n} \varphi\left(\frac{n}{d}\right) C(d),$$

with Euler totient φ .

Reflections. For even n , a half-length transfer with symmetry boundary conditions gives $R(n)$, the number of reflection-fixed sequences; for odd n , $R(n) = 0$. The reflection contribution is $\frac{n}{2}R(n)$.

Orbit count.

$$a(n) = \frac{\text{RotSum}(n) + \text{RefTerm}(n)}{2n}, \quad \text{RefTerm}(n) = \frac{n}{2}R(n).$$

5 Worked example: $n = 8$

Divisors: 1, 2, 4, 8. Transfer powers give

$$C(1) = 2, \quad C(2) = 4, \quad C(4) = 16, \quad C(8) = 256.$$

Weighted by totient:

$$\text{RotSum}(8) = 4 \cdot 2 + 2 \cdot 4 + 1 \cdot 16 + 1 \cdot 256 = 288.$$

Reflection routine: $R(8) = 16$, so $\text{RefTerm}(8) = 4 \times 16 = 64$. Therefore

$$a(8) = \frac{288+64}{16} = 22.$$

6 Optional filters

Primitivity. Using Möbius inversion,

$$a_{\text{prim}}(n) = \sum_{d|n} \mu(d) a\left(\frac{n}{d}\right).$$

Forbidden subsequences. Canonical signatures from smaller n may be declared forbidden; a length- n signature is rejected if any cyclic substring matches a forbidden canonical representative. This path enumerates signatures directly (used for small n).

7 Validation and cost

For $n \leq 15$ I compare Burnside+transfer counts with brute-force canonical enumeration; they agree. Unconstrained counts match OEIS A053656 on the tested range. Runtime is driven by divisor iteration and fast exponentiation of the fixed 6-state matrix, so million-scale n is practical.

8 Outlook

Next steps include asymptotic constants under additional local rules and extending the forbidden-subsequence filter with transfer-based acceleration.